SET-A **University Roll No.**_____

Term I Examination, 2019-20
B. Tech. II Year (Even Semester)

Subject: English for Professional Purposes–II **Subject Code: BELH 0004**
Time: 60 Minutes **Max. Marks: 15**

Note: It's mandatory to cleary mention your set no in answer sheet.
ii. Write only the options (such as A, B, C etc.) in the answer sheet while answering Verbal Aptitude questions.

PART A - ORAL SKILLS **TIME: 30 MINUTES**

Section- A **0.5x4 = 2 Marks**

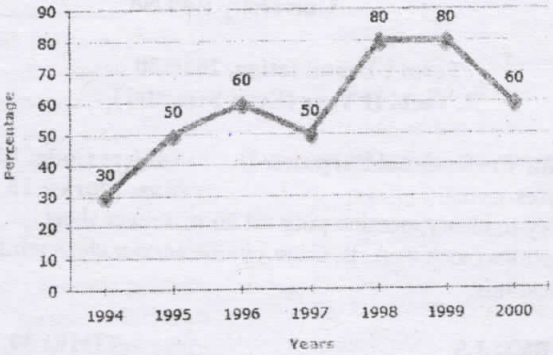
Note: Attempt all the questions.

1. What information does the last paragraph of a cover letter ideally contain?
2. Mention at least two differences between Curriculum Vitae and Resume.
3. Speaking is a combination of verbal and non-verbal means. True/False
4. The number of syllables in a word is equal to the number of the vowel sounds in that word. True/False

Section- B **1.5x2 = 3 Marks**

Note: Attempt any two questions. (In not more than 60 words)

5. Mark stress on the appropriate syllable in the underlined words and elaborate on the rule behind it.
   (i) They increase the cost of the blanket every year.
   (ii) There is some increase in the cost of the blanket.
   (iii) It is a wonderful object.
   (iv) I do not object to your plan.
   (v) We do not have any authority to permit you.
   (vi) Do you need a permit to work here?
6. You have worked with four different employers and now are applying for a new position as Mechanical Engineer in Tata Consultancy Service. Write your work experience in reverse chronological order.
7. The following line graph gives the percentage of the number of candidates who qualified an examination out of the total number who appeared for the examination over a period of seven years from 1994 to 2000. Interpret the data in your own words.



Section- C **2.5x1 = 2.5 Marks**

Note: Attempt any one question. (In not more than 125 words)

8. You are Gaurav Sharma. Recently you read an advertisement for the Software Engineer in Infosys Technologies Ltd., Banglore. Draft a cover letter to apply for the position.
9. Break the following words into syllables and mark stress on appropriate syllable.

Production,	Accent,	Politics,	Engineer,	Doctor,
Relative,	Photograph,	Democracy,	Contrast,	Accident

PART B- VERBAL APTITUDE TIME: 30 MINUTES

Note: Attempt all the questions. Each question carries 0.25 marks.

Mark the most appropriate choice as your answer. (Q 1-12)

1. He would have been awarded with an excellence certificate if he _____ extraordinarily.
   A. had performed
   B. are performing C. perform
   D. will perform E. performed
2. Do you think there would be less conflict in the world if all the people _____ the same language?
   A. would speak B. spoke
   C. were speaking D. speak
   E. would have spoken
3. If they _____hard, they will get good grades.
   A. worked B. will work
   C. would work
   D. would have worked
   E. are working
4. We would be nothing but a fool, if we _____ that opportunity.
   A. left B. would leave
   C. will leave D. leave
   E. had left
5. It _____ easy to paint pictures if she practised it before.
   A. would be B. will be
   C. would have been
   D. shall be
   E. be
6. If Dr Mehrotra ____ thirsty, he would have drunk some water.
   A. was B. has been C. had been
   D. would be E. will be
7. He availed himself _____ all the opportunities which life bestowed upon him.
   A. at B. for C. on
   D. to E. with

8. It is very cold and foggy weather. You must sit beside the fire for some time. The underlined word in the sentence is a _____.
   A. preposition B. adverb
   C. adjective D. conjunction
   E. none of these
9. The child insisted _____ watching a movie in the theatre.
   A. on B. of C. in D. over
   E. to
10. He went to Agra with a view _____ visiting the Taj Mahal.
   A. on B. of C. In D. for
   E. to
11. The Professor instructed the students to be _____ the campus for some more time.
   A. in B. on C. of D. over
   E. to
12. He is senior _____ Mr Gupta.
   A. to B. from C. than D. over
   E. none of these

Identify the erroneous part of the sentence. (Q 13-16)

13. Put you in my position (A)/ and you would realize (B)/ the problems faced(C)/ in my profession(D)/ No error (E)
14. The student requested the teacher (A)/ to explain him(B)/ the theory of relativity with some examples (C)/ which he could understand easily (D)/ No Error (E)
15. I must tell you that (A)/ I prefer (B)/ my job (C)/ to yours (D)/ No Error (E)
16. The work (A)/ of an uneducated farmer (B)/ is far important than (C)/ that of a professor (D)/ No Error (E)
17. He said, "Hurray! I won the match." The indirect speech of this sentence:
   A. He exclaimed with joy that he won the match.
   B. He exclaimed with joy that he had won the match.
   C. He was happy to win the match.
   D. He told joyfully that he would win the match.
   E. None of these
18. Which of the following statements about conditional sentences is not true?
   A. A conditional sentence is a complex sentence.
   B. A conditional sentence has at least two dependent clauses.
   C. Among the types of conditional sentences, first and second conditionals talk about future possibilities- one real and the other, unreal.
   D. Zero conditional describes scientific facts.
   E. A conditional clause does not necessarily begin with 'if'.
19. He described a superlative meal, the like of which he'd never eaten. Here 'the like' is used as:
   A. Noun
   B. Preposition
   C. Adverb
   D. Conjunction
   E. None of the above
20. When a verb has no object to receive its action, is called:
   A. A transitive verb
   B. An intransitive verb
   C. A finite verb
   D. A regular verb
   E. An auxiliary verb

Read the passage and answer the questions that follow. (Q 21-24)

If you are not in the grip of avarice, you will choose a trade or profession that appeals to you as the means of self-

expression and social service, even if you cannot earn much money by it. Your daily duty is not merely money-making drudgery; it is your contribution to social progress and the path of personal development. How sad must be the lot of the man or woman who must do uncongenial work simply for the sake of more money? I know a younger professor who loved literature passionately and taught it admirably; but his salary was small, and he became a lawyer in order to secure a larger income. I tell you that man was guilty of a crime, and he will not be happy as a lawyer, though he may be a little more comfortable, a little better fed and dressed and lodged. A musician who is born a violinist can never be personally happy or socially useful as a prosperous merchant or stock-broker, as he will miss the violin all the time. Avarice puts round men in square jobs and square men in rounds jobs. At present many persons in all classes, rich and poor, are unhappy and restless because their daily work is not interesting. It does not provide an outlet for the creative impulse and the insistent urge of personality. Therefore, don't ask in youth, 'How can I earn the biggest salary?' Ask rather, 'How can I be truly happy and do most good to society?' Then you may have less money, but you will have more of life and joy.

21. Which of the following is true?
    A. Avarice can make you earn a lot of money
    B. All those who choose a profession as a means of self-expression cannot earn money
    C. If a person is avaricious he/she will choose a profession that serves as a means of self-expression and social service
    D. Both A and B
    E. None of the above
22. Which of the following is not true?
    A. He did a crime
    B. His salary was small
    C. He was a little more comfortable
    D. He was a musician who was born a violinist
    E. None of the above
23. "Round men in square jobs and square men in rounds jobs" means:
    A. Difficult jobs
    B. Incompatibility
    C. Different kinds of jobs
    D. Both A and C
    E. None of the above
24. "According to the passage people are unhappy and restless because:
    A. Their daily work is not interesting
    B. They have less money
    C. They are guilty of a crime
    D. they want to a prosperous merchant or stock -broker
    E. All of the above

Write the meanings of the following idioms and frame sentences using them in a suitable context. (Q 25-30)

25. Play second fiddle
26. Dead wood
27. Worth one's salt
28. Get the axe
29. Waiting in the wings
30. Hot off the press

ME - 2nd yr.

Paper Code-BTDH 0302	SET-C

Soft Skills

First Mid Term, Even Sem Examination (2019-20)

Course- B.tech **Year- II**

Branch- (All)

Time : 60 Min. **M.M-20**

Instructions :

1. **Attempt all questions.**
2. **There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.**
3. **The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.**
4. **The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.**
5. **Only black& blue pen should be used to fill forms.**
6. **Each branch is identified with a specific subject code which should be written in subject column of OMR sheet**
7. **Write your University roll number in the roll no column of OMR sheet.**

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:..... **All The Best**

(Aptitude)-No of question-40, marks-0.50

Direction for (Q1-Q2): In each of the following questions, a word is represented by only one set of numbers. The columns and rows of a square matrix that have 36 cells are numbered from 0 to 5. A letter in a cell can be represented first by its column and then the row number.

	0	1	2	3	4	5
0	H	A	Z	N	X	P
1	C	G	P	H	S	G
2	N	S	C	V	A	W
3	Z	W	M	G	E	M
4	G	A	Y	O	Y	P
5	A	K	E	M	U	J

Q1. GAME:-

A. 04, 14, 23, 52 B.04, 10, 21, 43

C. 04, 10, 53, 43

D. 12, 10, 53, 43 E. None of these

Q2. HORSE:-

A. 00, 34, 31, 41, 25 B. 31, 34, 44, 41, 25

C. 31, 34, 40, 41, 25

D. Cannot be determined E. None of these

Q3. In a certain code language $#* means 'Shirt is clean', @ D# means 'Clean and neat' and @ ? means 'neat boy', then what is the code for 'and' in that language

A. # B. D C. @ D. Data inadequate E. None of these

Q4. In a certain code , RIPPLE is written as 613382 and LIFE is written as 8192. How is PILLER written in that code?

A. 318826 B. 776655 C. 786543 D. 156724

E. None of these

Q5. 'book' is coded as 'pencil', 'pencil' is coded as 'mirror', 'mirror is coded as 'book'. Then what is useful to write on a paper?

A. book B. pencil C. mirror D. Cannot be determined

E. None of these

Q6. In a certain code language 'CLOSE' is coded as 81, 'FAR' is coded as 56, then what is the code for AWAY
A. 18 B. 58 C. 86 D. 72 E. None of these

Q7. In a certain code language 'BREAK' is coded as 'pehtc', 'GLASS' is coded as 'xwdnh', then what is the code for SEVEN
A. sjxgt B. riygt C. rjxfs D. siygt E. None of these

Q8. In a certain code 'MISSIONS' is written as 'MSIISNOS'. How is 'ONLINE' written in that code?
A. OLNNIE B. ONILEN C. NOILEN D. LNOENI
E. ONNLIE

Q9. In a certain code 'ROAR' is written as 'URDU'. How is 'URDU' written in that code?
A. V X D Q B. XUGX C. ROAR D. VSOV
E. V Z C P

Q10. If the average number of 8 terms is given to be 40 and the average of first 6 terms is given to be 35.What is the average of the remaining 2 terms?
A. 30 B. 55 C. 40 D. 42 E. None of these

Q11. The average age of 14 girls and their teacher's is 15. If teacher's age is excluded the average reduces by 1. What is the teacher's age?
A. 35 years B. 32 years C. 30 years D. 29 years
E. None of these

Q12. A cricket batman had a certain average of runs for his 11 innings. In the 12th innings, he made a score of 90 runs and thereby his average of runs was decreased by 5. His average of runs after 12th innings:
A. 155 B. 150 C. 145 D. 140
E. None of these

Q13. The average monthly expenditure of a family is Rs. 2,200 during first three months. Rs. 2,550 during next four months and Rs. 3,120 during last five months of the years. If the total savings during the year was Rs. 1,260, what is the average monthly income?

A. Rs. 1,260 B. Rs. 1,280 C. Rs. 2,805
D. Rs. 2,850 E. None of these

Q14. The average of 6 observations is 45.5. If one new observation is added to the previous observations, then the new average becomes 47. The new observation is
A. 58 B. 56 C. 50 D. 46 E. None of these

Q15. The average of 7 consecutive numbers is n. If the next two numbers are included, the average will
A. increased by 2 B. remains the same C. increased by 1
D. increased by 2 E. None of these

Q16.Two alloys of iron having 85% and 67% iron respectively are fused together to form another alloy having 71% iron in it. If 14 kgs of the first alloy is taken, find the weight of the second alloy.
A. 28 kgs B. 34 kgs C. 49 kgs D. 42 kgs
E. None of these

Q17. How many litres of solutions having 30% milk and 70% milk should be mixed so that we get 40 litres of 45% milk solution?
A. 10 , 30 B. 12 , 28 C. 15 , 25 D. 25 , 15
E. None of these

Q18. Contents of three vessels having volumes in the ratio of 2 : 3 : 4 and having solutions with milk concentrations of 40%, 30% and 20% are mixed together in a fourth large vessel. Find the concentration of milk after the mixing.
A. 30% B. 27.7% C. 27.2% D. 26.1%
E. None of these

Q19. How many litres of water needs to be added to 25 litres of a solution having milk and water in the ratio 8 : 5, such that the resultant has milk and water in the ratio 5 : 8.
A. 15 B. 20 C. 25 D. 30 E. None of these

Q20. From a 100 litre container having milk and water in the ratio 8 : 7, 10 litres of solution is removed and replaced with 20 litres of milk. Again 20

litres of solution is removed and replaced with 30 litres of milk. Find the amount of milk in the solution now.
A. 112.5 B. 105.6 C. 93.7 D. 87.5 E. None of these

Q21. How many litres should be removed out of 40 litres of solution of milk and water and be replaced with equal amount of water so that the proportion of milk drops from 80% to 75%.
A. 2 lts B. 2.5 lts C. 3 lts D. 3.5 lts E. None of these

Q22. A 20 litre mixture of milk and water contains milk and water in the ratio 3 : 2. 10 litres of the mixture is removed and replaced with pure milk and the operation is repeated once more. At the end of the two removal and replacement, what is the ratio of milk and water in the resultant mixture?
A. 17 : 3 B. 9 : 1 C. 3 : 17 D. 5 : 3 E. None of these

Q23. A container contains 50 litres of milk. From this container, 10 litres of milk was taken out and replaced by water. This process is repeated one more time.
How much milk is now left in the container?
A. 24 B. 32 C.30 D.36 E. None of these

Q24. A container has 40 litres of wine from this container, 4 litres of wine is taken out and replaced with water. This process is repeated two or more times. What will be the final quantity of water in the container?
A. 22 B. 10.84 C. 15 D.20 E. None of these

Q25. A man sells two wrist watches at Rs 594 each. On one he gains 10% and on the other, he loses 10%. Find his gain or loss percent on the whole.
A. 1% loss B. 1% gain C. 3% loss D. Data is not sufficient
E. None of these

Q26. A shopkeeper sells a book at a discount of 10%. He earns a profit of 12%. Then the ratio of the C.P. to the marked price of the book is?
A. 33: 31 B. 3: 2 C. 45: 56 D. 2:3 E. None of these

Q27. Which of the following numbers is divisible by 99?
A. 114345 B. 913464 C. 135792 D. 3572404
E. None of these

Q28. If the price of petrol increases by 25% by how much % will a person reduce the quantity of petrol purchased, so that there will be no change in the overall expense?
A.16.67% B.25% C.20% D.33.33%
E.None of these

Q29. Ashish lent Rs. 5000 to Manish for 2 years and Rs. 3000 to Chintu for 4 years on simple interest at the same rate of interest and received Rs. 2200 in all from both of them as interest. The rate of interest per annum is:
A. 5% B. 7% C. 7.125% D. 10% E. None of these

Q30. In how many years Rs. 1000 becomes 1728 at the rate of 20% p.a.c.a. ?
A. 2 years B. 2.5 years C. 3 years D. 3.5 years
E. None of these

Q31. If 8th April, 2005 is Monday, what was the day of the week on 8th April, 2004?
A. Sunday B. Monday C. Thursday D. Saturday
E. None of these

Q32. An accurate clock shows 2 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 9 o'clock in the evening?
A. 144° B. 210° C. 168° D. 570°
E. None of these

Q33. The calendar for the year 2007 will be the same for the year?
A. 2014 B. 2016 C. 2018 D. 2017
E. None of these

Direction (Q34-35) -P and Q are brothers of R. Q is son of S and T. S is the daughter of U. A is the father-in-law of T. B is son of U.

Q34. What is the relationship of Q to B?
A. Nephew B. Neice C. Maternal Uncle D. Paternal Aunt
E. None of the above

Q35. What is the relationship of P to B?
A. Nephew B. Paternal Uncle C. Neice D. Paternal Aunt
E. None of the above

Q36. In a class of 40 children, Sunetra's rank is eight from the top. Sujit is five ranks below Sunetra. What is Sujit's rank from the bottom?
A. 27 B. 29 C. 28 D. 26 E. None of these

Q37. In a row of boys, A is fifteenth from the left and B is fourth from the right. There are three boys between A and B. How many boys are there in the row?
A. 9 B. 10 C. 14 D. 22 E. None of these

Q38. A, B and C play cricket. The ratio of the runs scored by A and B is 3 : 2. The ratio of the runs scored by B and C is 3 : 2. Together, they score 342 runs. How many runs did A make?
A. 162 B. 108 C. 72 D. Cannot be determined
E. None of these

Q39. A, B and C enter into a partnership by investing Rs.3000, Rs.4000 and Rs.5000. A is a working partner and gets a half of the profit for his services and the remaining profit is divided among the three in the rate of their investments. What is the amount of profit that B gets if A gets a total of Rs. 6000?
A. 1888.88 B. 1333.33 C. 2000 D. 1600
E. None of these

Q40. If Rs. 782 be divided into three parts, proportional to 1/2:2/3:3/4, then the first part is:
A. Rs. 182 B. Rs. 190 C. Rs. 196 D. Rs. 204
E. None of these

Printed Pages-2 **University Roll No.....**

First Mid-Term Examination, Even Semester 2019-20

B. Tech (CSE), II Year, IV Semester

BCSC0009: Software Engineering

Time: 01 Hours **Total Mark: 15**

Section-A

Note: Attempt all Questions **3*2=6 Marks**

I. What are the problems in the formulation of requirements in software development? Why it is difficult to improve software process.

II. A user placing an order with a sales company might follow these steps: 1. Browse catalog and select items, 2. Call sales representative, 3. Supply shipping information, 4. Supply payment information, 5. Receive conformation number from sales person. Consider the above problem of system and design the Use Case Diagram.

III. Consider a project with the following functional units:
Number of user inputs = 30
Number of user outputs = 42
Number of user enquiries = 08
Number of files = 07
Number of external interfaces = 6
Assume that value of complexity adjustment factors is 1.10, and weighting factors for all functional units are average. Compute the function points for the project. Also calculate size of project in LOC, if it is going to be developed in Java.

Section-B

Note: Attempt all Questions **3*3=9 Marks**

I. Draw Agile-Scrum framework. Name some Agile quality strategies. Explain the iterative and incremental development in Agile?

II. Consider Blood Bank Management system, which is designed to help the people in need of blood. This system is managing information of Blood, Blood group, Donor, Blood stock, Orders, Blood sells and Patient. Draw level-0 and level-1 DFD for this system.

III. Suppose a system for office automation is to be designed. It is clear from requirements that there will be five modules of size 500, 1500, 2000, 1000 and 2000 LOC respectively. The manager rates the project details as follows:

Characteristic	Level	EAF
Complexity	High	1.15
Reliability	High	1.15
Programmer's Capability	Low	1.17
Application Experience	Low	1.10

Other are consider as nominal. Use COCOMO model to compute the effort. Also find the number of people required for the computed effort.

Printed Pages: 01 **University Roll No:.....**

First Term Examination, Even Semester, 2019- 20

Programme: B-Tech **Branch:** CCV **Year:** II

Introduction Virtualization & Cloud Computing, BCSE 0502

Time: 1 Hour **Maximum Marks: 15**

Note: Attempt All the Questions

Section – A (3 x 2 = 6 Marks)

I. Explain how cloud computing has evolved from grid, cluster and utility computing?

II. Compare Traditional IT infrastructure and Virtualization with Cloud Computing.

III. Explain the structure of a cloud computing solution with the help of example.

Section – B (3 x 3 = 9 Marks)

I. Define Cloud Bursting? Also, what are the different solutions provided by hybrid cloud to avoid the problem of vendor lock-in?

II. It is challenging to equip an IT resource to match its processing requirements. If the demand for the IT resource is below its capacity, then it is under-utilized and if the demand is above its capacity it is over-utilized or unable to meet the demand. Develop a service model to show how can IT resources be scaled automatically in response to fluctuating demand?

III. Virtual servers run on a hypervisor, and hardware resources are emulated for the virtual servers via the hypervisors. If the hypervisor fails or if the underlying physical server fails (thereby causing the hypervisor to fail), the failure condition cascades to all of its hosted virtual servers. Design a cloud architecture to overcome this failure.

Printed Page: 2 University Roll No.....

First Mid-Term Examination, Even Semester 2019-20

Program: B-Tech- CSE, CCV, DA, CSF & IOT **Year:** II **Semester:** IV

BCSC 0008: Computer Networks

Time: 1 Hour **Maximum Marks: 15**

Note: Attempt All the Questions

Section – A 3 x 2 = 6 Marks

I. Differentiate OSI model and TCP/IP protocol suite. Enlist the layer which perform following services
   a) Route Determination
   b) Interface to transmission media
   c) Hop to Hop delivery
   d) Log in and Log out Email service

II. What are the causes of transmission impairment? Explain in brief. What will be the SNR value for the following
   a) Extreme Noisy Channel
   b) Noiseless Channel

III. (A) Explain in brief following terms
   a) Guided Media
   b) Unguided Media

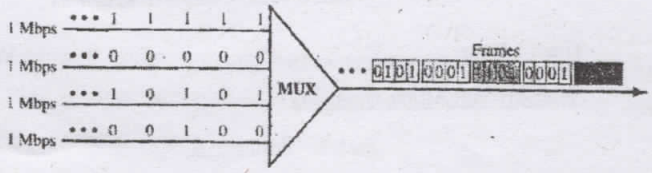
   (B) Draw Manchester and Differential Manchester graph for the bit stream 00000000 (eight consecutive zeros)

Section – B 3 x 3 = 9 Marks

I. The following figure shows synchronous TDM, with a data stream for each input and one output stream for the output. The unit of data is 1 bit. Find

   a) Input bit duration
   b) Output bit duration

c) Output bit rate (transmission rate of the link)
d) Output frame rate



OR

A path in a digital circuit switched network has a data rate of 1 Mbps (Mega bits per second). The exchange of 1000 bits are required for setup and teardown phase (connection establishment and connection release phase). The distance between both the parties is 5000 km. The propagation speed is $2 * 10^8$ m/s. What will be total delay if we transmit?

a) 1000 bits of data during data transfer phase
b) 1,00,000 bits of data during data transfer phase

II. Differentiate ALOHA and CSMA/CD. One hundred stations on pure ALOHA network shares a 1 Mbps channel. If the frames are 1000 bits long, find the throughput if each station is sending 10 frames per second.

III. What is the total delay for a frame size of 5 million (mega) bits that is being sent on a link with 10 routers each having queuing time of 2 microseconds and processing time of 1 microsecond. The length of the link is 2000 km. the speed of light inside the link is $2 * 10^8$ m/s. The link has bandwidth of 5 Mbps (mega bits per second). Which component of total delay is dominant? Which one is negligible?

Printed Pages: 01 University Roll No.....

First Mid-Term Examination, Even Semester 2019-20

B. Tech (CSE, IOT, DA, CCV & CSF), II, IV

BCSC 0007: Introduction to Microprocessor

Time: 1 Hour Maximum Marks: 15

Section – A

Attempt All the questions. **3 x 2 = 6 Marks**

1. Write any four features of 8085 microprocessor? List four applications of Microprocessor?
2. Draw and explain flag register of 8085 MPU. What will be the value of all flags (except auxiliary carry flag) of 8085 after performing subtract 8E by 7E hexadecimal data.
3. Write the working of following pins of 8085 MPU.

   I. HOLD and HLDA II. $\overline{\text{RESET IN}}$ and RESET OUT

Section – B

Attempt All the questions. **3 x 3 = 9 Marks**

1. Define Instruction cycle, machine cycle. The instruction MVI B, FF (hexadecimal) is useful to load FF (hexadecimal) as data byte in register B. It is a 2-Byte instruction. Draw the timing diagram for MVI B, FF (hexadecimal).
2. Write the assembly language program to subtract two sixteen bits numbers available in BC and DE register pair, after performing operation store the result in 3000 and 3001 memory location respectively. Also write description of each instruction with program.
3. List all addressing modes used in 8085 MPU. Explain any two with example.

Printed Pages: 1 University Roll No.....

First Mid-Term Examination, Even Semester 2019-20

B.Tech (CSE/CCV/DA/CSF/IoT), IInd, IVth

BCSC0006 : Data Structures & Algorithms

Time : 1 Hour Maximum Marks: 15

Section – A

3 x 2 = 6 Marks

1. Explain the benefits of using a Data Structure to store and process data over a standard array. Give an example.
2. Explain the linked-list data structure. Create a Linked-List Node class for a Doubly-Linked List to store the object of a Student class. Assume that the Student class is written for you.
3. What is postfix expression ? Write the postfix expression of

   (A – B) / ((C + D) * E).

Section – B

3 x 3 = 9 Marks

1. Examine the signature of the method given below

   ```
   public void printFibonacciSeries(int n) {
   // write your code here
   }
   ```

   Complete this method using recursion to print the first 'n' Fibonacci numbers. Call the method inside a sample Main class.

2. What do you understand by the term "Big O of ..."? What are Algorithmic Complexities? Arrange the most common known complexities in an ascending order.
3. Explain the "Last-In, First-Out" (LIFO) working principle of a Stack? Demonstrate a Stack with code using a Linked-List to store integers.

University Roll No.....

B.TECH. - II Year (IV-Semester)

First Term-I Exam (Even)

Title of Paper: Environmental Studies

Paper Code: (BCHS-0201)

Time: 01 Hours. **Max. Marks: 15**

Section – A

Note: Attempt All Questions. **[2x3]**

Ques1. Highlight the importance of studying environmental studies for acquiring sustainable environmental development. [2]

Ques2. Discuss the structural components of an ecosystem. [2]

Ques3. Suggest some methods for the conservation of mineral resources. [2]

Section – B

Note: Attempt All Questions. **[3x3]**

Ques1. With the help of a neat sketch classify various segments of atmosphere on the basis of temperature variation, highlighting important activities taking place in each segment. [3]

Ques2. How forests contribute for the economic and ecological importance of a country. [3]

Ques3. Write short notes on any **two (2)** of the following: [3]

A.) Food chain & its significance.

B.) Fluoride problem in drinking water.

C.) Types of natural resources.

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First Mid-Term Examination,
Even Semester 2019-20
B.Tech- II Year, IV-Semester
Operating System (BCSC0004)

Time: 1 Hour **Maximum Marks: 15**

Section – A

All question Carry equal marks: **3 x 2 = 6 Marks**

1. What are the differences among multiprogramming, multitasking and multiprocessing? Name the OS of each category.
2. Which scheduler decides the degree of multiprogramming and which scheduler swap out the partially executed processes to from ready queue to back up storage. Explain in Brief!
3. What are different states of process? In what conditions process switches from suspended ready and suspended waiting states.

Section – B

All question Carry equal marks: **3 x 3 = 9 Marks**

1. Calculate the response time of the process P3 for RR CPU Scheduling. (Where arrival time and burst time is in ms.) as per the following given data. Also calculate average waiting time. (Time Slice=3 ms)

PROCESS	ARRIVAL TIME	BURST TIME	PRIORITY
P1	0	8	3
P2	1	5	2
P3	2	2	1
P4	3	4	0

2. What are various types of kernels? Which type of kernel implements the drivers as a part of kernel (e.g. runs in kernel mode)? How micro kernel is beneficial over monolithic kernel?
3. Calculate the response time of the process P4 for Priority CPU Scheduling. (Where arrival time and burst time is in ms.) as per the data given in **Question 1 of Section-B**. Also calculate average waiting time. Clearly mention Starvation Problem and its solution with example. **[Priority : lower number gets higher priority 0>1>2>3]**

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First-Term Examination, Even Semester 2019-20

B. Tech (CSE), II Year, IV Semester

Subject & Code: Theory of Automata & Formal Languages (BCSC0011)

Time: 1 Hour Maximum Marks: 15

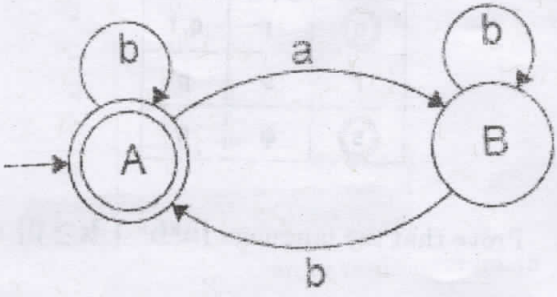
Section – A

Attempt all questions. [3 x 2 = 6 Marks]

A 1. Construct a DFA equivalent to the regular expression a* (ba*)*

A 2. Construct a Moore machine that takes the set of all strings over {0,1} as input and produces 'A' as output if the input ends with 10, produces 'B' as output if input ends with 11, otherwise produces 'C' as output.

A 3. Using Arden's Theorem, find the Regular Expression for the following DFA.



Section – B

Attempt all questions **[3 x 3 = 9 Marks]**

B 1. Construct minimum state automaton equivalent to the given transition table:

States/Inputs	0	1
→ q_0	q_1	q_0
q_1	q_0	q_2
q_2	q_3	q_1
(q_3)	q_3	q_0
q_4	q_3	q_5
q_5	q_6	q_4
q_6	q_5	q_6
q_7	q_6	q_3

B 2. Convert the NFA, whose transition table is given below, to equivalent DFA. **Only give the transition table. Clearly mark the initial and final states.**

	0	1
→p	q, s	q
(q)	r	q, r
r	s	p
(s)	φ	p

B 3. Prove that the language $\{a^k b^k \mid k \geq 0\}$ is not regular.

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First Term Examination, Even Semester 2019-20
B.Tech (DA), Year-II, Semester-IV

Subject Code : BCSE0552 **Subject Name: Applied Statistical Analysis**
Time : 1 Hours **Maximum Marks: 15**

Section- A

Note: Attempt All Three Questions. 3 x 2 =06 Marks

I. How do we measure qualitative and quantitative data? What is the difference between Nominal and Ordinal Scaling?

II. Find x and y so that the ordered data set has a mean of 42 and a median of 35.

17, 22, 26, 29, 34, x, 42, 67, 70, y

III. The following table shows the grouped data, in classes, for the heights of 50 people.

height (in cm) - classes	120-130	130-140	140-150	150-160	160-170
frequency	2	5	25	10	8

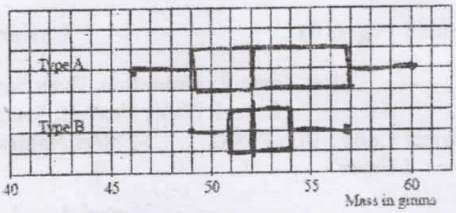
a) Calculate the mean of the salaries of the 20 people.

b) Calculate the standard deviation of the salaries of the 20 people.

Section – B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

I. A gardener collected data on two types of tomato. The box and whisker plot below shows data for the masses in grams of the tomatoes in the two samples. Compare and contrast the two types and advise the gardener which type of tomato he should grow in future.



II. (i) The probability that the political party A does a particular work is 30% and the political party B doing the same work is 40%. Then find the probability that the work is completed if the probability of choosing the political party A is 40% and that of B is 60%

(ii) Husband and wife apply for two vacant spots in a company. If the probability of wife getting selected and husband getting selected are 3/7 and 2/3 respectively, what is the probability that neither of them will be selected?

III. High Performance Bicycle Products Company in Chapel Hill, Mathura, sampled its shipping records for a certain day with these results

Time from receipt of order to Delivery *in Days)																			
4	12	8	14	11	6	7	13	13	11	11	20	5	19	10	15	24	7	29	6

Construct a frequency distribution for these data and a relative frequency distribution. Use intervals of 6 days.

(i) What statement can you make about the effectiveness of order processing from the frequency distribution?

(ii) If the company wants to ensure that half of its deliveries are made in 10 or fewer days, can you determine from the frequency distribution whether they have reached this goal.

(iii) What does having a relative frequency distribution permit you to do with the data that is difficult to do with only a frequency distribution?

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First Term Examination, Even Semester 2019-20
B.Tech (CSE), Year-II, Semester-IV
Subject Code: BCSC0010 **Subject Name: Discrete Mathematics**
Time: 1 Hours **Maximum Marks: 15**

Section- A

Note: Attempt All Three Questions. 3 x 2 =06 Marks

I. (i) Distinguish between ϕ, $\{\phi\}$, $\{0\}$, 0.
(ii) Prove the following:
$$A \cap B = A-(A-B) = B-(B-A)$$

II. Find the solution of recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 5.2^n$

III. In a group of 70 persons 37 like coffee, 52 like tea and each person like at least one of the two drinks calculate (i) How many like both coffee and tea (ii) How many like coffee but not tea.

Section – B

Note: Attempt All Three Questions. 3 x 3 = 9 Marks

I. Let A = {1, 2, 3, ..., 11, 12}, and let R be a relation on A x A defined by $(a,b)R(c,d)$ if and only if $a+d=b+c$. Prove that R is an equivalence relation.

II. Use generating function to solve the recurrence relation
$$a_n - 9a_{n-1} + 20a_{n-2} = 0 \quad a_0 = -3, a_1 = -10$$

III. (i) In how many ways can a committee of 5 teachers and 4 students be chosen from 9 teachers and 15 students?
(ii) Show that if any 20 people are selected then we may choose a subset of 3 so that all 3 were born on the same day of the week.

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First Mid-Term Examination, Even Semester 2019-20

B.Tech(CSE)-CSF, II Year, IV Semester

BCSE0602: IT Network Security

Time : 1 Hour **Maximum Marks: 15**

Section – A

Note: All questions are compulsory **3 x 2 = 6 Marks**

Q1. Discuss different types of Network attacks? Highlight the basic difference between the Trojan horse and Spyware?

Q2. What is Social engineering? What type of access attack can result if someone does the Social engineering? Explain. How ARP poisoning can be prevented.

Q3. A company is granted the site address 166.86.0.0. The company needs 1800 subnets. Show the complete process of designing the subnets in the given scenario.

Section – B

Note: All questions are compulsory **3 x 3 = 9 Marks**

Q1. Briefly explain basic functionalities of all the seven layers of OSI. Also explain how it is associated with TCP/IP model.

Q2. Discuss the two security protocols which can be used to provide security at transport layer. What is the basic difference between PGP and S/MIME?

Q3. Show the complete process of Digital Signature for message verification. Which encryption technique is used in Digital Signature? What is the best way of distributing public keys over the network?

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First- Term Examination

Even Semester, 2019-20

Programme: B-Tech Branch: IoT Year: II

Cloud Architecture & Deployment Models, BCSE 0652

Time: 1 Hour Maximum Marks: 15

Note: Attempt All the Questions

Section – A (3 x 2 = 6 Marks)

I. Compare desktop virtualization with application virtualization?

II. Discuss the shortcomings of traditional IT infrastructure. Also, compare Virtualization with traditional IT infrastructure?

III. Suggest a fast storage infrastructure to an organization that has offices in different cities. Support your solution with proper reasoning. Also, Compare above storage solution with NAS.

Section – B (3 x 3 = 9 Marks)

I. Compare full virtualization with para virtualization? Also, compare hardware assisted virtualization with software assisted virtualization.

II. When a virtual server is moved from one physical host to another, it loses its original network configurations, resulting in loss of network connectivity until the original configurations are recreated on the destination host. Suggest a network model using this network loss can be recreated.

III. Identify the types of virtualization, that an organization should use, that is planning to set up its various offices in different part of the world. Justify your answer with proper reasoning. Also, discuss the cost impacts of proposed virtualizations.

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First Term Examination, Even Semester 2019-20
Programme: B.Tech Year: II Sem:IV
Database Management Systems: BCSC 0003

Time: 1 Hour **Maximum Marks:15**

Section – A

3 x 2 = 6 Marks

I. List four significant differences between a file-processing system and a DBMS.

II. What is the importance of data independence in the context of DBMS? Explain with suitable examples.

III. Enlist various categories of End Users in DBMS.

Section – B

3 x 3 = 9 Marks

I. Construct an ER Diagram for Company having following details:
- Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations.
- A department controls a number of PROJECT. Projects have a unique name, number and a single location.
- Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by her dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee.
- Employee's DEPENDENT are tracked for health insurance purposes (dependent name, birthdate, relationship to employee).

II. What are the different levels of abstraction in the DBMS? Explain each level in detail.

III. Give the following queries in the relational algebra using the relational schema:

student(id, name)
enrolledIn(id, code)
subject(code, lecturer)

a) What are the names of students enrolled in cs3020?
b) What are the names of students in both cs1500 and cs1200?
c) What are the names of students taking a subject taught by Roger.